

AYUSH SUKHADAN NAIR

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Education

Fr. C. Rodrigues Institute of Technology

Bachelor of Engineering in Information Technology

- CGPA: 9.52/10.0

Vashi, Navi Mumbai

2022 – 2026

Work Experience

Tata Institute of Fundamental Research (TIFR)

Project Intern

Colaba, Mumbai

June 2025 – Present

- Engineered an end-to-end **automated wirebond inspection pipeline** deployed on **Linux (Ubuntu)**, using heatmap correlation for orientation correction, ORB-RANSAC homography for coordinate registration, and CLAHE + Lanczos upscaling for stephole extraction across 92 sites per board.
- Integrated a **custom RF-DETR object detection model** via **REST API** for gold pad inspection, with network timeout handling, per-stephole confidence scoring, and automated PDF/CSV QC report generation.
- Launched **HexaVision**, a **PyQt5 desktop application** with multi-threaded pipeline, real-time progress tracking, and stephole grid viewer; deployed on **Linux** using **venv**.
- Awarded **1st prize** in poster presentation for this project at PCUBE.

Sports Mentoring Infusion - NGO

Remote

Backend + Analytics Developer

June 2025 – July 2025

- Delivered a production-grade application with **base authentication system** and **all core modules** implemented.
- Architected platform infrastructure to serve **50+ users** across **4 user roles** (admins, students, coaches, volunteers).
- Streamlined communication workflows to reduce manual coordination time by **70%** for **10+ coaches**.

Indian Institute of Science Education and Research (IISER)

Mohali, Punjab

Project Intern

July 2024 – June 2025

- Engineered fall detection system using **YOLOv8-OB** + **MediaPipe**, reducing false positives by **60%**.
- Created healthcare solution for **live monitoring and feedback** for immediate response.

Fr. Conceicao Rodrigues Institute of Technology

Vashi, Navi Mumbai

Developer Intern

July 2023 – June 2024

- Optimized college portal serving **200+ users** with enhanced performance and functionality.
- Designed an information management system for managing college data and information.

Projects

Delta Investments | MERN Stack, AI Integration

2024

- Architected investment platform with **real-time stock data** and **Gemini AI** integration.
- Integrated an **auto trading feature** with AI-powered market analysis and risk management.

VaniConnect: Sign Language Transcriber | Python, Computer Vision

2024

- Constructed real-time gesture translation system with **85% accuracy** across **200+ sign patterns** for both **Indian and American sign languages**.
- **Won 1st place** in Summer Project Competition among **60+ participants**.

Open Source Contribution Tracker | React, Node.js, MongoDB

2024

- Constructed an analytics platform to provide analysis and tracking for **20+ GitHub repositories**.
- Enabled users to view details of their open-source contributions and generate shareable links to showcase them on platforms like LinkedIn.

Achievements & Leadership

Horizon 1.0 Hackathon – OnCampus GDSC, Vidyavardhini's College of Engineering & Technology Vasai, India

1st Place Winner

March 2026

- Secured **1st place** by building a **data aggregation platform** for SMEs that transforms multi-vertical business data into **real-time insights, alerts, and a live dashboard** for informed decision-making.

GirlScript Summer of Code (GSSOC'25)

Open Source Program

Selected Contributor

2025

- Contributed to **AlgoVisualizer** project by adding **custom array input feature** for sorting and searching techniques.
- Enhanced **frontend design** of documentation page, improving user experience and accessibility.

HackQuinox 2024 - CSI FCRIT

Mumbai, India

Top 5 Finalist

2024

Computer Society of India - IT Department

FCRIT

Treasurer

2023

- Handled finance for CSI-IT events, managing budgets up to **INR 30,000** with sponsorship incentives.

Technical Skills

Languages: Python, Java, JavaScript, SQL

Frontend: React, HTML/CSS, JavaScript

Backend: Node.js, Express, MongoDB

ML/AI: TensorFlow, Scikit-learn, Computer Vision

Databases: MongoDB, MySQL, ChromaDB (Vector Database)

Tools: Docker, GitHub Actions